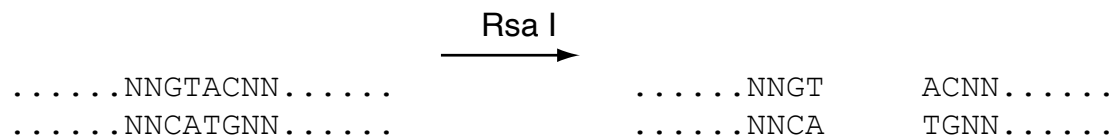
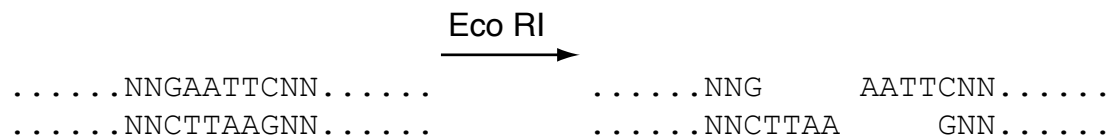
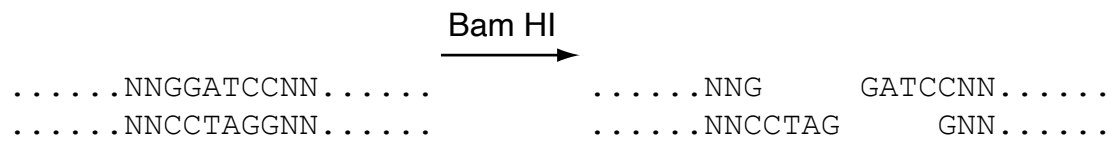


LAB 2: GENOME STRUCTURE

Restriction endonucleases are enzymes that cut defined sequences of DNA. Some examples are shown below.



Are these sites distributed equally in the genome? Is the distribution the same for all enzymes and organisms?

Working Materials

Mealworm beetle (*Tenebrio molitor*)

Mouse (*Mus musculus*) liver

Bacteria (*Escherichia coli*)

Restriction endonucleases Eco RI, Bam HI and Rsa I

Procedures

Genomic DNA extraction

1. Sample preparation

1.1. Bacteria

- 1.1.1. Add 1.5 ml of stationary phase bacterial culture to a microcentrifuge tube.
- 1.1.2. Centrifuge bacteria for 1 min at 4,000rpm.
- 1.1.3. Remove and discard all supernatant.
- 1.1.4. Resuspend pelleted bacteria in 50 μ l of resuspension buffer (RBuff).
- 1.1.5. Add 10 μ l of 10 mg/ml lysozyme (Lys).
- 1.1.6. Put on ice for 5 min. Proceed to 2.

1.2. Tissues

- 1.2.1. Cool tube with ball bearings in liquid nitrogen.
- 1.2.2. Add 0.05 g tissue to tube
- 1.2.3. Shake to pulverise tissue. Keep tube cold.
- 1.2.4. Put on ice until sample starts to melt.
- 1.2.5. Proceed to 2.

2. Extraction

- 2.1. Add 300 μ l of DNA Lysis buffer (DNA Lys) and mix gently but well.
- 2.2. Incubate at 65°C overnight.
- 2.3. Cool the samples on the bench.
- 2.4. Add 1 μ l of 5 μ g/ μ l RNase A (RNAgone) to the sample and mix well.
- 2.5. Incubate at 37°C for 30 min.
- 2.6. Place the samples on ice for 3-5 min.
- 2.7. Add 150 μ l of Protein Precipitation Reagent (Hi Salt) and mix gently but well.
- 2.8. Place the samples on ice for 3 minutes.
- 2.9. Centrifugation for 10 min at 4°C at top speed.
- 2.10. Transfer the supernatant to a new microcentrifuge tube. Discard the pellet.
- 2.11. Add 500 μ l of isopropanol (IsoP) to the supernatant. Mix by inverting tube many times (>20).
- 2.12. Centrifuge for 10 min at 4°C at top speed.
- 2.13. Remove the isopropanol. Wait 2 min and carefully remove the remaining isopropanol.
- 2.14. Rinse the pellet carefully with 70% ethanol (70% EtOH). Remove all ethanol.
- 2.15. Leave the open tube on the side on the bench to evaporate remaining ethanol.
- 2.16. Resuspend the DNA in 50 μ l of Tris/EDTA Buffer (TE). Store.

Restriction enzyme digestion of genomic DNA

1. In a microfuge tube
2. Add 11 μ l of ddH₂O
3. Add 6 μ l of DNA
4. Add 2.0 μ l of 10X Restriction enzyme buffer (Use the right buffer for your enzyme!)
5. Add 1 μ l of restriction enzyme (either *Eco*RI or *Bam*HI, or *Rsa*I)
6. Incubate in 37° C water bath for chosen time (up to 60 minutes)
7. Place sample on ice
8. Add DNA gel loading buffer (1/9 volume of sample)
9. Mix well with a pipette tip

Agarose gel electrophoresis

1. Make sure electrodes are not connected

HAZARD: high voltage

2. Put samples into the bottom of the wells
3. Attach electrodes from power supply to tank
4. Remember: DNA moves from negative (black) to positive (red)
5. Set to 100 Volts.